

Q1. Choose the correct answer, then fill up the table with correct choice:

The exam starts here

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Points 2

1. What does product quality depend on?
☒ a. Development process
b. Developer skills
c. Project size
d. All of the above
2. What is one of the process attributes that focuses on how easy it is to understand the process definition?
a. Standardization
b. Understandability
c. Visibility
d. Measurability
3. Which of the following is a key principle of Extreme Programming (XP)?
a. Long-term planning
b. Continuous feedback
c. Heavy documentation
d. Heavy documentation
4. In Extreme Programming (XP), what is the practice of continuously improving the code?
☒ a. Refactoring
b. Pair Programming
c. Test Automation
d. Continuous Integration

1	2	3	4
a	b	b	a

Q2. According to the 8 principles of the IEEE Code of Ethics (PUBLIC, CLIENT AND EMPLOYER, PRODUCT, JUDGMENT, MANAGEMENT, PROFESSION, COLLEAGUES, SELF), identify the type of ethics reflected in the following descriptions:

Points/4

STATE THE EXAMPLE	ETHICS PRINCIPLE
Give a fair hearing to the opinions, concerns, or complaints of a colleague.	Public
Improve their knowledge of this Code, its interpretation, and its application to their work.	Self
Ensure that software engineers are informed of standards before being held to them.	Management
Ensure an appropriate method is used for any project on which they work or propose to work.	Product
Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.	Management
Maintain the integrity of data, being sensitive to outdated or flawed occurrences.	Product
Accept full responsibility for their own work.	Professional
Not give unfair treatment to anyone because of any irrelevant prejudices.	Public
	Self

Q3. Extract 2 functional requirements and 2 non-functional requirements from the following software description.

Points/4

A company needs a Task Management System to streamline its internal processes. The system will allow employees to create tasks with a title, description, priority level, and due date, and assign these tasks to members. Employees can update the status of their tasks as they work on them. The system should handle 100 concurrent users without delays and ensure that user data is encrypted during transmission.

- Functional
1. The system will allow employees to create tasks.
 2. Employees can update the status of their tasks.
- Non-functional
1. The system should handle up to 100 concurrent users without delays.
 2. ensure that user data is encrypted during transmission.

Q4. What are the 4 main software engineering activities. The activities must be given in a logical order to ensure the successful delivery of software products:

- 1) Software Specification
- 2) Development
- 3) Verification and Validation
- 4) Documentation

Points: 2

Q5. Select the most appropriate software development model (Plan-Driven, Iterative, or Agile) based on key project characteristics. Justify your choice.

Points: 4

1. Project 1 Description:

- **Stakeholder Presence:** Limited. Stakeholders provide initial requirements but are not available for frequent feedback.
- **Requirement Volatility:** Low. Requirements are well-defined and unlikely to change.
- **Project Size:** Large-scale project with multiple teams working in parallel.
- **Team Structure:** Teams are geographically distributed, and communication is primarily through documentation.

Development Model: Plan Driven

Justification:

in Plan Driven all details Specification and design
before implements so it's not available
frequent feedback - uses the water fall
which is inflexible and each
depend on its previous work in

2. Project 2 Description:

- **Stakeholder Presence:** High. Stakeholders are actively involved and provide frequent feedback.
- **Requirement Volatility:** High. Requirements are expected to change as the project progresses.
- **Project Size:** Small team of 5 developers working on a web application.
- **Team Structure:** Co-located team with daily stand-up meetings.

Development Model: Agile

Justification: Agreement on the Principles of the ARIEL Model
 The "Contract is ~~unilateral~~ Involved"
 "Progress or Regret" which means ~~the~~ Yacenta ~~the~~
early feedback from the community and the ~~the~~ is
not as that ability?

3. **Project 3 Description:**

- **Stakeholder Presence:** Moderate. Stakeholders are available for periodic reviews but not continuously.
- **Requirement Volatility:** Moderate. Some changes are expected, but the core requirements are stable.
- **Project Size:** Medium-sized project with a team of 15 developers.
- **Team Structure:** The team is co-located, and there is a need for regular iterations to refine the product.

- **Stakeholder Presence:** Moderate. Stakeholders are available for periodic reviews but not continuously.
- **Requirement Volatility:** Moderate. Some changes are expected, but the core requirements are stable.
- **Project Size:** Medium-sized project with a team of 15 developers.
- **Team Structure:** The team is co-located, and there is a need for regular iterations to refine the product.

Development Model: iterative involves the customer work with medium project
Justification: incremental can ~~involve~~ the customer work with large project
the size of the project is larger than what all can handle
and we can do some changes in the iterative model
~~"can solve the problem like plan driven"~~

Development Model: iterative
Justification: incremental can ~~not~~ ^{involve} the customer work with medium project
the size of the project is larger than what agile can handle
and we can do some changes in the iterative model
~~it can solve the problem like the plan driven~~

4. **Project 4 Description:**

- **Stakeholder Presence:** Limited. Stakeholders provide initial requirements and review progress at major milestones.
- **Requirement Volatility:** Low. Requirements are stable and well-documented.
- **Project Size:** Medium-sized project with a focus on regulatory compliance.
- **Team Structure:** Teams are experienced and follow a structured process with clear documentation.

- Project 4 Description:**
- **Stakeholder Presence:** Limited. Stakeholders provide initial requirements and review progress at major milestones.
 - **Requirement Volatility:** Low. Requirements are stable and well-documented.
 - **Project Size:** Medium-sized project with a focus on regulatory compliance.
 - **Team Structure:** Teams are experienced and follow a structured process with clear documentation.

• Team Structure: Plan driven

Development Model: Plan driven

Justification: ~~Plan driven is not suitable for large and medium projects.~~

Plan driven work will work with large and medium projects.

Plan driven is inflexible to change.

4/4

Justification: ~~Plan driven work will~~ with large, P. and medium

Plan driven is unflexible to change